VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

Single-Phase RLC Series Circuit Impedance and Power Calculator

A9205 - BASIC ELECTRICAL ENGINEERING LABORATORY

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Abstract

The Single-Phase RLC Series Circuit Analyzer is a computational tool designed to evaluate key electrical parameters of an RLC series circuit under alternating current (AC) conditions. This project focuses on calculating impedance, current, power factor, and real power using user-provided inputs such as resistance (R), inductance (L), capacitance (C), supply voltage (V), and frequency (f).

The system computes inductive reactance and capacitive reactance to determine the overall impedance of the circuit, which governs the current flow. Based on these values, the power factor and power consumption are evaluated, helping in understanding the efficiency of the circuit. Additionally, the project highlights the concept of resonance, where inductive and capacitive reactances become equal, resulting in minimum impedance and maximum current.

This project is implemented using a programming-based approach, providing accurate and instant results, making it useful for students and engineers in analyzing AC circuits. It serves as an educational tool to bridge theoretical concepts with practical computation, enhancing the understanding of fundamental electrical engineering principles.

Keywords:RLC Series Circuit, Impedance, Power Factor, Resonance, AC Circuit Analysis, Electrical Power

1 Introduction

Electrical circuits consisting of resistance (R), inductance (L), and capacitance (C) are fundamental to the study of alternating current (AC) systems. Among these, the single-phase RLC series circuit plays a crucial role in understanding the behavior of electrical parameters such as impedance, current, and power factor under varying frequency conditions. These circuits are widely used in practical applications including power systems, communication devices, and electronic equipment.

In an RLC series circuit, resistance opposes current flow, inductance stores energy in the form of a magnetic field, and capacitance stores energy in an electric field. The combined effect of inductive and capacitive reactance determines the overall impedance of the circuit, which directly influences the current and power consumption. The phase difference between voltage and current further introduces the concept of power factor, an important parameter that indicates the efficiency of power utilization.

This project aims to develop a computational tool that analyzes a single-phase RLC series circuit by calculating key parameters such as inductive reactance, capacitive reactance, impedance, current, power factor, and power. It also emphasizes the phenomenon of resonance, where the inductive and capacitive reactances become equal, resulting in minimum impedance and maximum current.

By integrating theoretical concepts with a programming-based approach, this project provides a simple and efficient method to analyze AC circuits. It helps in enhancing conceptual understanding while offering practical insights into real-world electrical systems.

2 Literature Review

The analysis of single-phase RLC series circuits has been a fundamental topic in electrical engineering, widely discussed in standard textbooks and academic research. Basic concepts such as impedance, reactance, and power factor are well established in classical works like Electrical Engineering Fundamentals by Vincent Del Toro and Engineering Circuit Analysis by William H. Hayt. These resources provide detailed theoretical foundations for understanding AC circuit behavior under different conditions.

Research studies and educational tools have emphasized the importance of computational methods for circuit analysis. With the advancement of programming and simulation technologies, software-based approaches have become increasingly popular for analyzing RLC circuits. Tools such as MATLAB and Multisim allow users to simulate circuit behavior, visualize waveforms, and validate theoretical results. These platforms demonstrate how computational techniques can simplify complex calculations and improve accuracy.

Several studies have also focused on the significance of resonance in RLC circuits. Resonance occurs when inductive and capacitive reactances are equal, resulting in minimum impedance and maximum current. This concept is widely applied in communication systems, filter circuits, and tuning applications. Literature highlights that understanding resonance is essential for designing efficient and stable electrical systems.

In recent years, there has been a shift toward developing simple, user-friendly tools for educational purposes. Many projects and applications aim to create calculators and interactive systems that help students easily compute circuit parameters without manual calculations. These tools bridge the gap between theoretical knowledge and practical implementation, making learning more effective.

This project builds upon these existing studies by developing a simplified computational tool specifically focused on single-phase RLC series circuit analysis. It combines theoretical principles with a programming-based approach to provide accurate, quick, and user-friendly results, contributing to both academic learning and practical understanding.

3 Objectives

- To develop a computational tool for analyzing RLC circuits
- To calculate impedance, current, power factor, and power
- To reduce manual calculation errors
- To analyze the effect of frequency
- To provide a user-friendly system

4 Theory Background

A single-phase RLC series circuit consists of a resistor (R), inductor (L), and capacitor (C) connected in series to an AC supply. In such circuits, resistance opposes current, inductance stores energy in a magnetic field, and capacitance stores energy in an electric field.

In AC circuits, opposition to current is called impedance (Z), which is a combination of resistance and reactance. Reactance is of two types: inductive reactance (XL) and capacitive reactance (XC).The relationship between voltage and current depends on the difference between XLand XC, which determines whether the circuit behaves as inductive or capacitive. The phase difference between voltage and current leads to the concept of power factor, which indicates the efficiency of power usage.

At resonance, XL=XC, resulting in minimum impedance and maximum current.

5 Mathematical Formulation

$$
\begin{aligned}
X_L &= 2\pi f L \\
X_C &= \frac{1}{2\pi f C} \\
Z &= \sqrt{R^2 + (X_L - X_C)^2} \\
I &= \frac{V}{Z} \\
\cos\phi &= \frac{R}{Z} \\
P &= V I \cos\phi
\end{aligned}
$$

6 Algorithm

1. Start
2. Input R, L, C, V, f
3. Calculate X_L and X_C
4. Compute impedance Z
5. Calculate current I
6. Calculate power factor
7. Calculate power
8. Display results
9. Stop

7 Program Code

```
import numpy as np
import matplotlib.pyplot as plt

# ----------- USER INPUT -----------
R = float(input("Enter Resistance (Ohm): "))
L = float(input("Enter Inductance (H): "))
C = float(input("Enter Capacitance (F): "))
V = float(input("Enter Voltage (V): "))

# Frequency range for graph
f = np.linspace(1, 500, 500)  # 1 Hz to 500 Hz

# ----------- CALCULATIONS -----------

XL = 2 * np.pi * f * L
XC = 1 / (2 * np.pi * f * C)

Z = np.sqrt(R**2 + (XL - XC)**2)
```

```
I = V / Z
pf = R / Z

# ----------- SINGLE VALUE OUTPUT (at 50 Hz) -----------

f0 = 50

XL0 = 2 * np.pi * f0 * L
XC0 = 1 / (2 * np.pi * f0 * C)
Z0 = np.sqrt(R**2 + (XL0 - XC0)**2)
I0 = V / Z0
pf0 = R / Z0
P0 = V * I0 * pf0

print("\n--- Results at 50 Hz ---")
print(f"XL = {XL0:.2f} Ohm")
print(f"XC = {XC0:.2f} Ohm")
print(f"Z = {Z0:.2f} Ohm")
print(f"Current = {I0:.2f} A")
print(f"Power Factor = {pf0:.3f}")
print(f"Power = {P0:.2f} W")

# ----------- GRAPHS -----------

# 1. Frequency vs Impedance
plt.figure()
plt.plot(f, Z)
plt.xlabel("Frequency (Hz)")
plt.ylabel("Impedance (Ohm)")
plt.title("Frequency vs Impedance")
plt.grid()

# 2. Frequency vs Current
plt.figure()
plt.plot(f, I)
plt.xlabel("Frequency (Hz)")
plt.ylabel("Current (A)")
plt.title("Frequency vs Current")
plt.grid()

# 3. Frequency vs Power Factor
plt.figure()
plt.plot(f, pf)
plt.xlabel("Frequency (Hz)")
plt.ylabel("Power Factor")
plt.title("Frequency vs Power Factor")
plt.grid()

plt.show()
```

8 Sample Input and Output

Input: R=10, L=0.1, C=100μF, f=50Hz, V=230V

Output:

- $X_L = 31.4\Omega$
- $X_C = 31.8\Omega$
- $Z = 10.03\Omega$
- $I = 22.93A$
- Power Factor ≈ 0.997
- $P \approx 5260W$

9 Graphs

Frequency vs Impedance and Frequency vs Current graphs can be plotted to observe resonance behavior.

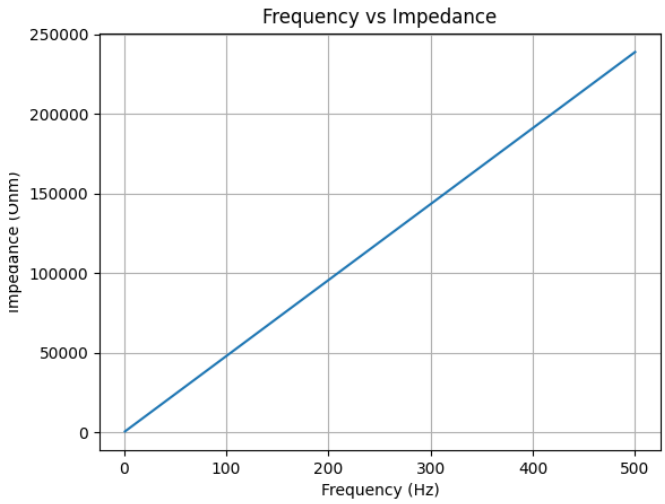

Figure 1: Frequency vs Impedance

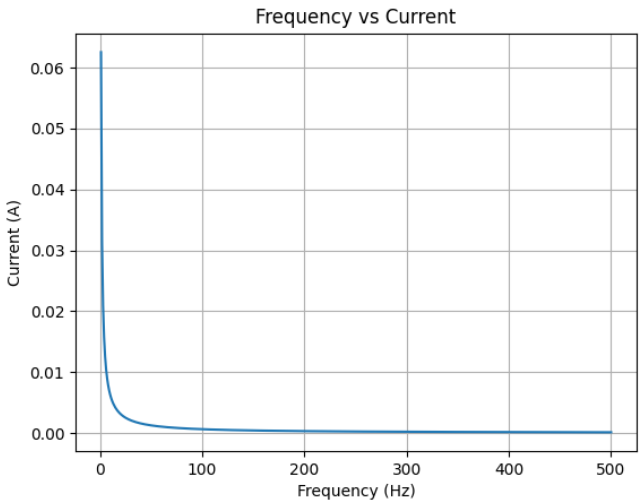

Figure 2: Frequency vs Current

10 Result Analysis

The results show that impedance varies with frequency. At resonance, impedance is minimum and current is maximum. The calculator provides accurate and efficient analysis of circuit parameters.

11 Conclusion

The project successfully analyzes a single-phase RLC series circuit using a computational approach. It simplifies calculations and improves understanding of AC circuit concepts such as impedance, power factor, and resonance.

12 References

- Engineering Circuit Analysis
- Electrical Engineering Fundamentals
- Basic Electrical Engineering

13 Mapping POs and SDGs

13.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Single Impedance Analyser

PO No.	**Program Outcome**	**Relevance to Project**
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, and engineering fundamentals to develop solutions for complex engineering problems.	Applied mathematical formulations of impedance ($Z = \sqrt{R^2 + (X_L - X_C)^2}$), reactance, and power factor using core circuit theory.
PO2	**Problem Analysis:** Identify, formulate, and analyse complex engineering problems reaching substantiated conclusions.	Identified the need for automated impedance calculation, formulated the RLC circuit problem, and derived correct conclusions from computed electrical parameters.
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems to meet identified needs.	Designed a C++ program that computes X_L, X_C, Z, I, $\cos\phi$, and P from user-supplied values of R, L, C, V, and f.
PO4	**Conduct Investigations:** Use research-based knowledge including modelling, analysis, and interpretation of data to provide valid conclusions.	Validated computed results against theoretical RLC analysis, modelling the effect of varying R, L, C, and frequency f on overall impedance.
PO5	**Modern Tool Usage:** Select and apply appropriate techniques and modern engineering and IT tools, recognising their limitations.	Utilised C++ with `<cmath>` library functions (`sqrt`, `pow`) to implement the analyser; compiled and tested using a standard IDE.
PO9	**Communication:** Communicate effectively within the engineering community, including writing effective reports and design documentation.	Results are displayed with clearly labelled output variables and the project is documented with structured code and this report.
PO11	**Life-Long Learning:** Recognise the need for independent learning and adaptability to new and emerging technologies.	The project encourages exploration of signal processing, filter design, and embedded systems as natural extensions of the impedance analysis concept.

13.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Single Impedance Analyser

SDG No.	**Goal**	**Relevance to Project**
SDG 4	Quality Education	Provides a practical computational tool for students to understand RLC circuit behaviour, reinforcing engineering education through hands-on software implementation.
SDG 7	Affordable and Clean Energy	Accurate impedance and power factor ($\cos\phi$) analysis supports power system efficiency. Improved power factor reduces reactive power losses, contributing to cleaner and more affordable energy use.
SDG 9	Industry, Innovation, and Infrastructure	Promotes innovation in electrical engineering tools by demonstrating how software-based analysers can replace manual calculations in industrial circuit design and testing workflows.
SDG 12	Responsible Consumption and Production	Software-based analysis reduces dependence on physical test instruments in early design stages, lowering resource consumption and encouraging digital engineering practices.
SDG 17	Partnerships for the Goals	Well-documented open code promotes knowledge sharing among students, educators, and engineers, supporting collaborative and inclusive technical development.